
The Coordinated Attack among Many Robots

Bystanders to a Lossy Message, the Cost of Each Level
of Mutual Knowledge, and Four Robots on the Floor

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The companion report states common knowledge of a stand as the precondition of a joint lift by two robots, proves that a lossy radio cannot supply it, and executes the policy a beacon makes possible. This report writes the same domain for N robots and m message levels. Every robot takes a share of the load, so the lift requires common knowledge over all N , and a message at level ℓ states that its sender knows that everyone knows, $\ell - 1$ times over, which stand the order names.

With more than two robots a message has bystanders, robots that neither send nor receive it. We give them an observability type of their own, relating the three events of the message to one another, and show that the product update with such a message still preserves S5 and still adds at most one level of mutual knowledge, so that no finite exchange yields common knowledge, for any number of robots. A signal observed from viewpoints that are common knowledge yields it over all N in one update. The planner returns the beacon policy, of depth $N + 3$, for up to ten robots, and its proofs on the radio floor run out of budget from three robots and two levels. Among N robots it found E^k to cost $k(N - 1)$ messages in every cell it settled, and showed in the others that fewer do not suffice: a message says more than its content, because a robot that knows the stand can only have learnt it from one that knew first. Four robots were executed in Gazebo through ePlanSys. Over the radio, six delivered messages reached E^2 and the executor refused the lift; with the beacon, one signal gave common knowledge over all four, and the four lifted together, arrivals 0.10s apart.

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1 Scope

The companion report [5] concerns two robots, south and north, which must lift one load together after one of them has read which of two stands it is on. Its results are stated for the group of the two, and two of its arguments use that the group has two members: the proof that a message adds at most one level of mutual knowledge distinguishes only the sender and the receiver, and the floor has one viewpoint at each mouth of the bay. This report removes the restriction. The domain is written for any number N of robots, all of which take a share of the load, and for any number m of message levels, and the question is which of the companion report's results survive, and at what cost to the planner.

Four claims are advanced.

1. When a robot that neither sends nor receives a message relates its three events, the lossy message preserves S5 and adds at most one level of mutual knowledge in a group of any size (Propositions 1 and 2). On the radio floor no policy exists, for any number of robots or messages (Corollary 1).
2. One signal from viewpoints that are common knowledge yields common knowledge over all N (Proposition 3), and the planner returns a policy of depth $N + 3$ for every N up to ten, each of whose leaves the product update confirms.
3. Among N robots the fewest messages that reach E^k were $k(N - 1)$ in every cell the planner settled, and at least $k(N - 1)$ in the others. The protocols it finds use what a message implies beyond its content.
4. Four robots executed both floors in simulation, with the precondition checked by the executor against the model, and the depth of mutual knowledge computed from the live model after each message agreed with the offline replay.

Four bounds are stated at the outset. The search settles the radio floor only for two robots up to five levels and for three robots at one level; beyond that it ran out of budget, and the floor is settled by Corollary 1 and not by search. The count $k(N - 1)$ is a pattern over the cells, minimal in each within the family of messages the domain defines, and is not proved in general. The floor has places for four robots, so the runs stop at four, while the planner's tables go to ten. And the positions domain of the companion report, in which a move is witnessed only by the robot that makes it, is written for a pair and was not extended. The bounds of the companion report also hold here: every message in the runs is delivered, the lift is drawn by moving the load in the simulator, and the positions of the robots are common knowledge because the move to a viewpoint is public.

2 Preliminaries

The definitions are those of the companion report, restated in the form this report uses.

Definition 1 (Epistemic model). Let $\mathcal{A}$ be a finite set of agents and P a finite set of atoms. An *epistemic model* is $\mathcal{M} = (W, R, V, W_d)$ with W a finite set of worlds, $R : \mathcal{A} \rightarrow 2^{W \times W}$, $V : W \rightarrow 2^P$, and $\emptyset \neq W_d \subseteq W$ the designated worlds. $\mathcal{M}$ is S5 when every R_a is an equivalence. A formula holds at $\mathcal{M}$ when it holds at every designated world.

Definition 2 (Group modalities). For $G \subseteq \mathcal{A}$ let $R_G = \bigcup_{a \in G} R_a$ and R_G^+ its transitive closure. $\mathcal{M}, w \models E_G \varphi$ iff φ holds at every v with $w R_G v$; $E_G^{k+1} \varphi = E_G E_G^k \varphi$ and $E_G^0 \varphi = \varphi$; and $\mathcal{M}, w \models C_G \varphi$ iff φ holds at every v with $w R_G^+ v$.

Definition 3 (Distance and depth). For $\mathcal{M} \models \varphi$, $\text{dist}_{\mathcal{M}}(\varphi)$ is the least m such that some designated world reaches, by an R_G -path of length m , a world where φ fails, and ∞ if there is none. The *depth* of φ is $\text{depth}_{\mathcal{M}}(\varphi) = \text{dist}_{\mathcal{M}}(\varphi) - 1$.

Lemma 1. If $\mathcal{M}$ is S5 and $\mathcal{M} \models \varphi$, then for $k \geq 1$, $\mathcal{M} \models E_G^k \varphi$ iff $k \leq \text{depth}_{\mathcal{M}}(\varphi)$, and $\mathcal{M} \models C_G \varphi$ iff $\text{dist}_{\mathcal{M}}(\varphi) = \infty$.

Proof. $E_G^k \varphi$ holds at w iff φ holds at every world reached from w by an R_G -path of length exactly k . Each R_a is reflexive, so the worlds reached in exactly k steps are those reached in at most k , and $E_G^k \varphi$ fails at a designated world iff a counterexample lies within distance k of it. The case of C_G is the union over k . $\square$

Definition 4 (Event model and product update). An *event model* is $\mathcal{E} = (E, T, Q, \text{pre}, \text{post}, E_d)$ with events E , observability types T , $Q : T \rightarrow 2^{E \times E}$, preconditions, postconditions and designated events E_d . Each agent a has a type τ_a for the whole model, the type whose observability condition holds at the designated worlds. The product update $\mathcal{M} \otimes \mathcal{E}$ has worlds $W' = \{(w, e) : \mathcal{M}, w \models \text{pre}(e)\}$, relations $(w, e) R'_a (v, f)$ iff $w R_a v$ and $(e, f) \in Q(\tau_a)$, valuation by the postconditions, and designated worlds $W'_d = (W_d \times E_d) \cap W'$. $\mathcal{E}$ is *applicable* in $\mathcal{M}$ when every designated world has a designated event whose precondition holds there.

Definition 5 (Lossy message). For agents $i \neq j$, a formula π and an atom p , $L(i, j, \pi, p)$ has events d (delivered), l (lost) and n (never sent), $\text{pre}(d) = \text{pre}(l) = \pi$, $\text{pre}(n) = \top$, $\text{post}(d) = \text{post}(l) = \{p \mapsto \top\}$, $\text{post}(n)$ trivial, $E_d = \{d\}$, and types

$$Q(\text{Sender}) = \{d, l\}^2 \cup \{(n, n)\}, \quad Q(\text{Receiver}) = \{(d, d)\} \cup \{l, n\}^2,$$

with $\tau_i = \text{Sender}$ and $\tau_j = \text{Receiver}$.

The sender knows it sent and cannot tell whether the message arrived; the receiver knows whether something arrived and, when nothing did, cannot tell a loss from silence. With two robots every agent is the sender or the receiver. With more, the definition leaves the other robots without a type.

3 The domain for N robots

The generator `tools/scaled.py` writes the domain for N robots and m message levels, and the launch file runs it with `robots:=` and `messages:=`. The robots are south, north, south2, north2, and so on, alternately on the storage and the dispatch floor; G is the set of all of them, and south reads the order. Every robot takes a share of the load, so $\text{lift}(s)$ requires $C_G \text{job}(s)$. A message at level ℓ from i to j is the lossy message $L(i, j, \pi, p)$ with

$$\pi = K_i E_G^{\ell-1} \text{job}(s) \wedge \neg p, \quad p = \text{sent}_{\ell}(i, j) :$$

the sender knows that everyone knows, $\ell - 1$ times over, which stand the order names, and each level goes from each sender to each receiver at most once. Level 1 is *tell*, level 2 *ack*, level 3 *ack2*, and so on. Table 1 gives the actions.

Table 1: The actions of the domain for N robots and m levels.

action	type	events	observers
$\text{read-order}(i, s)$	semi-private sensing	$e_h: \text{job}(s); e_e: \neg \text{job}(s)$; both with $\neg \text{at-view}(i) \wedge \neg K w_i \text{job}(s)$	i fully, every other robot partially
$\text{tell}, \text{ack}, \dots (i, j, s)$, level $\ell \leq m$	lossy message	$d, l: K_i E_G^{\ell-1} \text{job}(s) \wedge \neg \text{sent}_{\ell}(i, j)$, post $\text{sent}_{\ell}(i, j)$; $n: \top$	i sender, j receiver, every other robot a bystander
$\text{go-view}(i)$ $\text{signal}(i, s)$	public ontic conditional announcement	$\text{beacon} \wedge \neg \text{at-view}(i)$, post $\text{at-view}(i)$ $\sigma: \text{at-view}(i) \wedge K_i \text{job}(s)$; $n: \top$	everyone at a viewpoint fully, elsewhere obviously
$\text{lift}(s)$	public ontic	$\text{job}(s) \wedge C_G \text{job}(s)$, post lifted	everyone

Two robots. With $N = 2$ the content of each message is the companion report's. E_G is then $K_{south} \wedge K_{north}$, and $K_i E_G^{\ell-1} job(s)$ is equivalent in S5 to the alternating $K_i K_j K_i \dots job(s)$ with ℓ operators, the conjuncts that repeat an agent collapsing by introspection. The log differs. The companion domain lets each level go out once in all, and this one once per sender and receiver, so that at $N = 2$ it admits each message and its mirror image. Its beacon floor has the policy of depth five, found in 137 expansions where the companion domain takes 115; its radio floor is exhausted at depth 9 with 2 224 expansions, where the companion's is exhausted at depth 5 with 34.

4 Robots that neither send nor receive

Definition 6 (Bystander). In a lossy message $L(i, j, \pi, p)$ among a group $G \supseteq \{i, j\}$, every $b \in G \setminus \{i, j\}$ has $\tau_b = \text{Bystander}$, with $Q(\text{Bystander}) = E \times E$.

A robot that does not hear the radio cannot tell a delivery from a loss or from silence. The library's Oblivious type, which maps every event onto n , would not serve. Since d and l set the sender's log, an oblivious bystander at (w, d) would consider possible only worlds at which p fails, so that R'_b would not be reflexive: the bystander would believe that no message went out when one did, and the model would leave S5. The type is added to `epddl/lossy.epddl` (Listing 1). The companion report's floors never assign it, and their grounded tasks are unchanged apart from the added relation.

```
(:action-type lossy-message
:events (?del ?lost ?nil)
:observability-types (Sender Receiver Bystander Oblivious)
:relations (Sender (:and (?del ?del) (?del ?lost) (?lost ?del) (?lost ?lost) (?nil ?nil))
Receiver (:and (?del ?del) (?lost ?lost) (?lost ?nil) (?nil ?lost) (?nil ?nil))
Bystander (:forall (?e ?f - event) (?e ?f))
Oblivious (:forall (?e - event) (?e ?nil)) )
:designated (?del)
:conditions (?nil (:trivial-event)))
```

Listing 1: The action type, in `epddl/lossy.epddl`, with the bystander type.

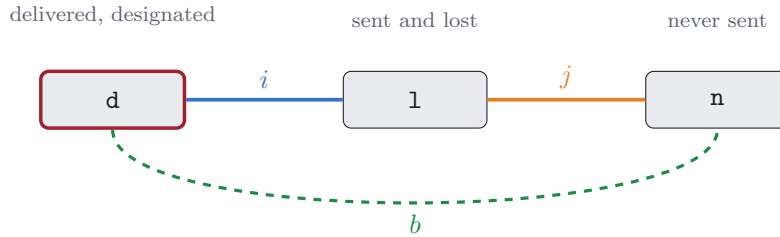


Figure 1: The lossy message among three or more robots. Reflexive edges are omitted. The sender i relates d and l , the receiver j relates l and n , and a bystander b relates all three events; only the edge it adds between d and n is drawn.

Proposition 1 (S5 is preserved). If $\mathcal{M}$ is S5, so is $\mathcal{M} \otimes L(i, j, \pi, p)$, with any number of bystanders.

Proof. R'_a is the restriction to W' of the relation $\{((w, e), (v, f)) : w R_a v, (e, f) \in Q(\tau_a)\}$ on $W \times E$. For each of the three types $Q(\tau_a)$ is an equivalence, $E \times E$ among them, so this is a product of equivalences, and its restriction to a subset is an equivalence. $\square$

Proposition 2 (At most one level per message, in any group). Let G be a finite group with $i \neq j$ in G and every other member a bystander of $L = L(i, j, \pi, p)$. Let $\mathcal{M}$ be S5 with $\mathcal{M} \models \varphi$, let φ not mention p , and let π be such that $\mathcal{M}, w \models \pi$ and $w R_i v$ imply $\mathcal{M}, v \models \pi$. If L is applicable in $\mathcal{M}$, then

$$\text{dist}_{\mathcal{M} \otimes L}(\varphi) \leq \text{dist}_{\mathcal{M}}(\varphi) + 1.$$

Proof. If $\text{dist}_{\mathcal{M}}(\varphi) = \infty$ there is nothing to show. Otherwise let $w \in W_d$ and $w = v_0 \xrightarrow{a_1} v_1 \xrightarrow{a_2} \dots \xrightarrow{a_m} v_m$ be an R_G -path with $m = \text{dist}_{\mathcal{M}}(\varphi) \geq 1$ and $\mathcal{M}, v_m \not\models \varphi$. Since L is applicable, $(w, d) \in W'_d$. Since $\text{pre}(n) = \top$, $(v, n) \in W'$ for every $v \in W$, and since $(n, n) \in Q(\tau)$ for all three types, $vR_a u$ implies $(v, n)R'_a(u, n)$ for every $a \in G$: the n -copies form a copy of $\mathcal{M}$. As $\text{post}(n)$ is trivial and φ does not mention p , (v_m, n) falsifies φ . Three cases.

$a_1 = j$. $(v_0, l) \in W'$ since $\text{pre}(l) = \text{pre}(d)$. Then $(v_0, d) R'_i(v_0, l)$ by reflexivity and $(d, l) \in Q(\text{Sender})$, and $(v_0, l) R'_j(v_1, n)$ by $v_0 R_j v_1$ and $(l, n) \in Q(\text{Receiver})$: two steps.

$a_1 = i$. $\mathcal{M}, v_1 \models \pi$ by the hypothesis on π , so $(v_1, l) \in W'$. Then $(v_0, d) R'_i(v_1, l)$ by $v_0 R_i v_1$ and $(d, l) \in Q(\text{Sender})$, and $(v_1, l) R'_j(v_1, n)$ by reflexivity and $(l, n) \in Q(\text{Receiver})$: two steps.

$a_1 = b$, a bystander. $(v_0, d) R'_b(v_1, n)$ by $v_0 R_b v_1$ and $(d, n) \in Q(\text{Bystander})$: one step.

In each case at most two steps reach (v_1, n) , and $m - 1$ further steps along the n -copy of the path reach (v_m, n) : a path of length at most $m + 1$ from a designated world to a counterexample. $\square$

The first two cases are the companion report's proof. The third is the only one the bystanders add, and it shortens the path: a bystander's edge from the delivered event to the event in which nothing was sent is an edge of the product the two-robot proof had to build in two steps. The proof does not use that d is the event that occurred, so the bound holds whether the message is delivered or lost.

Corollary 1 (No finite exchange yields common knowledge). Under the hypotheses of Proposition 2, if $\mathcal{M} \not\models C_G \varphi$ then $\mathcal{M} \otimes L_1 \otimes \dots \otimes L_k \not\models C_G \varphi$ for every finite sequence of lossy messages. On the radio floor, for every $N \geq 2$ and every m , no sequence of applicable actions makes $\text{lift}(s)$ applicable for either s , and no policy for lifted exists.

Proof. The first part is Propositions 1 and 2 and Lemma 1, by induction on k . On the radio floor *beacon* is false, so *go-view* and *signal* are never applicable. A message requires $K_{\text{south}} \text{job}(s)$ or a statement over it, which fails until the order is read, and *read-order* is applicable at most once. Every execution is therefore the reading followed by lossy messages. Before the reading, $\text{job}(s)$ fails at one of the two designated worlds. After it, on the branch where the order names s , $\text{job}(s)$ holds at the designated world, and every robot but south, observing the reading partially, relates it to the copy of the other world: $\text{dist}(\text{job}(s)) = 1$. The hypothesis on π holds for every message. $K_i E_G^{\ell-1} \text{job}(s)$ is preserved along R_i by positive introspection, and $\text{sent}_{\ell}(i, j)$ is set only by a message that i sends, whose Sender type separates n from d and l , so that its value is the same at all worlds R_i relates. By the first part the distance stays finite after every message, and by Lemma 1 $C_G \text{job}(s)$ never holds. $\square$

5 The beacon for N robots

Proposition 3 (A signal seen from common viewpoints). Let $\mathcal{M}$ be S5 with *at-view*(a) true at every world of $\mathcal{M}$ for every $a \in G$. If $S = S(i, K_i \chi)$ is applicable in $\mathcal{M}$, then $\mathcal{M} \otimes S \models C_G \chi$, whatever the size of G .

Proof. Every type is Fully, so $(w, e)R'_a(v, f)$ implies $e = f$, and every world R_G^+ -reachable from a designated (w, σ) is of the form (v, σ) . $(v, \sigma) \in W'$ requires $\mathcal{M}, v \models K_i \chi$, which gives $\mathcal{M}, v \models \chi$; as $\text{post}(\sigma)$ is trivial, χ holds at every reachable world. $\square$

As in the companion report, *at-view* is set by the public *go-view*, so once every robot has moved its position is common knowledge, and the hypothesis holds. The policy the planner returns is the companion report's with a viewpoint for every robot: each robot but the reader goes to its viewpoint, the order is read, the reader goes to its viewpoint and signals, and the robots lift. Its depth is $N + 3$, and it has two leaves, one per stand. The product update of the trace tool replays every policy found and reports $C_G \text{job}(s)$ at both leaves, for every N up to ten.

6 Planning results

Tables 2 and 3 are written by `tools/scaling.py`: plank grounds each cell, and Aletheia solves it at one thread, once with AO* and once replanning over the all-outcomes determinization, each with a budget of 300 s. AO* deepens one action at a time and checks its budget inside each iteration. An entry *cut off at depth d* means that the budget ran out in the iteration at depth d , so that no policy of smaller depth exists. The beacon floor is written with four message levels throughout, which it never uses.

Table 2: The beacon floor with N robots: the depth of the policy, the expansions and the time. Every policy found has two leaves, and the trace tool reports $C_G\text{job}(s)$ at both. Atom counts are plank’s, and include the log atoms $\text{sent}_\ell(i, i)$, which no action sets.

N	atoms	actions	AO*	replanning
2	24	28	depth 5, 137, 0.0 s	depth 5, 8, 0.0 s
3	46	65	depth 6, 868, 0.2 s	depth 6, 11, 0.0 s
4	76	118	depth 7, 8 038, 4.6 s	depth 7, 14, 0.0 s
5	114	187	depth 8, 108 298, 112.3 s	depth 8, 17, 0.0 s
6	160	272	cut off at depth 8, 165 772	depth 9, 20, 0.1 s
7	214	373	cut off at depth 7, 99 175	depth 10, 23, 0.1 s
8	276	490	cut off at depth 7, 84 110	depth 11, 26, 0.2 s
9	346	623	cut off at depth 7, 78 032	depth 12, 29, 0.4 s
10	424	772	cut off at depth 7, 68 467	depth 13, 32, 0.7 s

AO* pays about a factor of ten in expansions for each robot. At six robots its budget runs out one iteration short of the policy, and from seven on it does not reach depth eight. Replanning finds the same policy in $3N + 2$ expansions, ten robots in under a second.

Table 3: The radio floor with N robots and m levels. There is no policy; a cell gives how the search ended, the expansions and the time.

N	m	atoms	actions	AO*	replanning
2	1	12	16	exhausted at depth 3, 14, 0.0 s	refuted, 5, 0.0 s
2	2	16	20	exhausted at depth 5, 52, 0.0 s	refuted, 15, 0.0 s
2	3	20	24	exhausted at depth 7, 282, 0.1 s	refuted, 81, 0.0 s
2	4	24	28	exhausted at depth 9, 2 224, 1.7 s	refuted, 563, 0.8 s
2	5	28	32	exhausted at depth 11, 18 990, 45.9 s	refuted, 3 989, 18.4 s
2	6	32	36	cut off at depth 12, 89 529	cut off, 16 350
3	1	19	29	exhausted at depth 7, 294, 0.1 s	refuted, 67, 0.1 s
3	2	28	41	cut off at depth 10, 36 517	cut off, 353
3	3	37	53	cut off at depth 10, 74 958	cut off, 26
4	1	28	46	cut off at depth 9, 20 500	cut off, 435
4	2	44	70	cut off at depth 9, 48 601	cut off, 17
5	1	39	67	cut off at depth 8, 37 629	cut off, 15

On the radio floor what is measured is the price of proving that no policy exists. AO* proves it by exhausting the space. Replanning proves it by refuting the determinization, which is sound for unsolvability, since any branch of a policy of the task is a plan of its determinization. At two robots both proofs finish up to five levels, and each level costs AO* a factor that rises to about eight. A third robot leaves one level within reach, and a fourth none: the space is every order in which up to $N(N - 1)m$ messages can go out, and it outgrows the search long before the robots fill the floor. Where a search was cut off, nothing is claimed from it; that none of these floors has a policy is Corollary 1.

How the counts were taken. Aletheia, at commit 314bc29, prints a search’s expansions only when it finds a policy. The counts for searches that ended without one were taken with a build that also prints them there and changes nothing else; on the four floors of the companion report it returns the installed planner’s results, expansion for expansion. Replanning overran its budget on the larger radio floors, in a step that does not check the clock: by 40 % at five robots, and in an earlier run, at four robots and two levels, to 673 s before it was stopped. The harness kills a run at one and a half times its budget and thirty seconds more. Eight searches ran at once, alongside four processes of the search of Section 7, on sixteen hardware threads, so the expansions are the measure and the times are orders of magnitude.

7 What each level costs

`tools/ladder.py` gives the planner the radio floor with the goal $E_G^k \text{job}(s)$ in place of *lifted*. AO* deepens one action at a time, so the first policy it finds has the fewest messages that reach E_G^k , among the messages of the domain. The trace tool’s product update then replays the policy; it must agree on the depth after every message, and find at the end that $C_G \text{job}(s)$ fails and *lift* is not applicable.

Table 4: The fewest messages that reach $E_G^k \text{job}(s)$ among N robots. An entry $\geq c$ is a search cut off in the iteration that would hold c messages: no protocol of fewer exists, and none of c was found. The budget was 600 s, and an hour for four robots and E_G^3 . Blank cells were not run.

N	E_G^1	E_G^2	E_G^3	E_G^4	E_G^5
2	1	2	3	4	5
3	2	4	6	8	≥ 10
4	3	6	≥ 9		
5	4	≥ 8			

Wherever the search finished, E_G^k cost $k(N - 1)$ messages, and wherever it was cut off it had shown that fewer do not suffice. For two robots this is the bound of Proposition 2, attained. For more, the count is a pattern over the cells, minimal in each within this family of messages, and it is not proved in general.

It is fewer messages than their levels suggest. A robot that knows where the load is can only have learnt it from one that knew first, and the messages of the history are the only ones that can have occurred, so a message says more than its content: when north tells south2 $K_{\text{north}} \text{job}(s)$, south2 also learns that south knew, since north could have heard it from no one else. The protocols the planner finds use this. With three robots they relay along a line, south, north, south2, and back. With four, E_G^2 is a relay out to north2 and a broadcast back from it (Table 5), and this is the protocol the mission runs on the four-robot radio floor.

Table 5: The protocol for E_G^2 among four robots, on the branch where the order names s_1 . Worlds are those of the model the epistemic state kept during the run; the offline replay gives the same depths and counts.

action	$ W $	depth of $\text{job}(s_1)$
$\text{read-order}(\text{south}, s_1) \rightarrow e_h$	2	E^0
$\text{tell}(\text{south}, \text{north}, s_1)$	4	E^0
$\text{tell}(\text{north}, \text{south2}, s_1)$	6	E^0
$\text{tell}(\text{south2}, \text{north2}, s_1)$	8	E^1
$\text{ack}(\text{north2}, \text{north}, s_1)$	10	E^1
$\text{ack}(\text{north2}, \text{south2}, s_1)$	16	E^1
$\text{ack}(\text{north2}, \text{south}, s_1)$	34	E^2
$\text{lift}(s_1)$	not applicable: $C_G \text{job}(s_1)$ fails	

The third message is the first to raise the depth. After it north2 knows the stand, and also that south2 knew it, that north did, and that south did, since each could have learnt it only from the one before; that is E_G^1 , and each robot knows it of every other only once north2's acknowledgements have gone back down the line.

8 The floor with four robots

The floor is the companion report's: the pass-through floor, a racking block across a 30×50 m hall, the loads in bays t_1 and t_3 as the stands s_1 and s_2 , and the beacon in the open bay t_2 (Figure 2). south2 starts at the east end of the storage floor's southern aisle and north2 at the west end of the dispatch floor's northern aisle. Each mouth of t_2 holds two viewpoints, and under a load each robot takes one corner of its end.

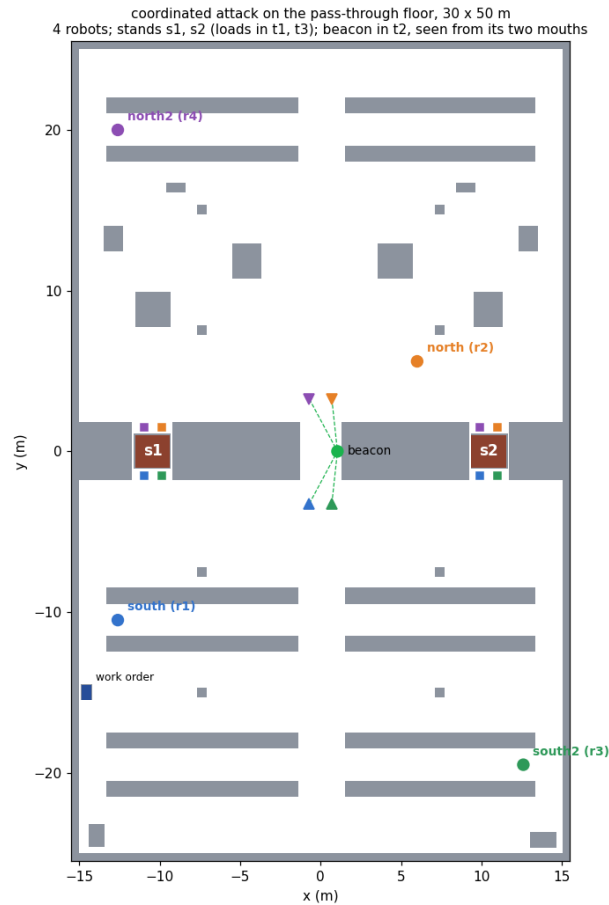


Figure 2: The floor plan with four robots: south and south2 on the storage floor, north and north2 on the dispatch floor, two viewpoints at each mouth of t_2 , and four corners under each load.

The floor generator checks the four geometric claims of the companion report for every robot and every pair, by casting rays over the occupied cells of the plan: no robot starts in sight of another or of the beacon; from its viewpoint each robot sees the beacon; no robot under one end of a load sees one under the other end; and every place a robot is sent is reachable over the inflated plan. Two robots under the same end of a load see each other, which is the same side, so meeting at the stand still gives no common knowledge over the four.

The performers are those of the companion report, made general. The signal node checks that every robot the model places at a viewpoint has the beacon in line of sight. The radio node puts the message on the receiver's inbox and prints its content at any level, $K_i E_G^{\ell-1} job(s)$. For *lift* every robot drives to its mouth of the stand, one start time is fixed for all, each drives in under its corner along the line from its mouth, and the load is raised.

9 Runs

Both floors were run with four robots and two message levels, the order naming s_1 , and recorded on a 3840×1080 virtual display with RViz on the left and Gazebo on the right. The beacon floor was also run with the order naming s_2 , without recording. In the tables the model column gives worlds and designated worlds after the update, and the depth of mutual knowledge of $job(s_1)$ among the four.

9.1 The radio floor

Table 6: The radio floor with four robots, as the run reported it.

action	reported	model
plan	no policy for <i>lifted</i> after 120.2s; cut off at depth 8	2 / 2
<i>read-order</i>	at the terminal after 13s; the order names s_1 ; e_h	2 / 1, E^0
<i>tell(south, north)</i>	" $K_{south}job(s_1)$ " delivered	4 / 1, E^0
<i>tell(north, south2)</i>	" $K_{north}job(s_1)$ " delivered	6 / 1, E^0
<i>tell(south2, north2)</i>	" $K_{south2}job(s_1)$ " delivered	8 / 1, E^1
<i>ack(north2, north)</i>	" $K_{north2}Ejob(s_1)$ " delivered	10 / 1, E^1
<i>ack(north2, south2)</i>	delivered	16 / 1, E^1
<i>ack(north2, south)</i>	delivered	34 / 1, E^2
<i>lift(s_1)</i>	knowledge requirement does not hold: (C (south north south2 north2) job_s1)	refused

On this floor the planner's 120s are a budget and not a proof: it was cut off in the iteration at depth 8, and the mission then ran the protocol of Table 5. The depths computed from the live model after each message were those the replay computes offline, and the mission confirmed through the epistemic state that $E_G^2job(s_1)$ held among the four robots and $C_Gjob(s_1)$ did not. No robot moved toward a stand. The floor runs to E_G^2 , where the companion report's runs to E_G^4 , because the planner did not find the four-robot protocols for E_G^3 or E_G^4 within its budgets.

9.2 The beacon floor

Table 7: The beacon floor with four robots, as the run reported it.

action	reported	model
plan	10 policy items, 2 leaves, 3.5 s	2 / 2
<i>go-view(north)</i>	at its viewpoint after 17 s; beacon in line of sight, 3.3 m	2 / 2
<i>go-view(south2)</i>	at its viewpoint after 59 s; 3.4 m	2 / 2
<i>go-view(north2)</i>	at its viewpoint after 61 s; 3.9 m	2 / 2
<i>read-order</i>	at the terminal after 15 s; the order names s_1 ; e_h	2 / 1, E^0
<i>go-view(south)</i>	at its viewpoint after 55 s; 3.8 m	2 / 1, E^0
<i>signal(south, s_1)</i>	in line of sight: all four, 3.3 to 3.8 m	3 / 1, C
<i>lift(s_1)</i>	at the mouths after 26 to 28 s; joint start; under the load after 15.1 and 15.2 s: starts 0.00 s apart, arrivals 0.10 s apart; load raised 0.12 m, 51 s after the lift began	1 / 1, C

With the order naming s_2 the read gave e_e , the s_2 tier was lit with all four robots in line of sight, $C_Gjob(s_2)$ held, and the load on s_2 was raised with arrivals 0.70 s apart.

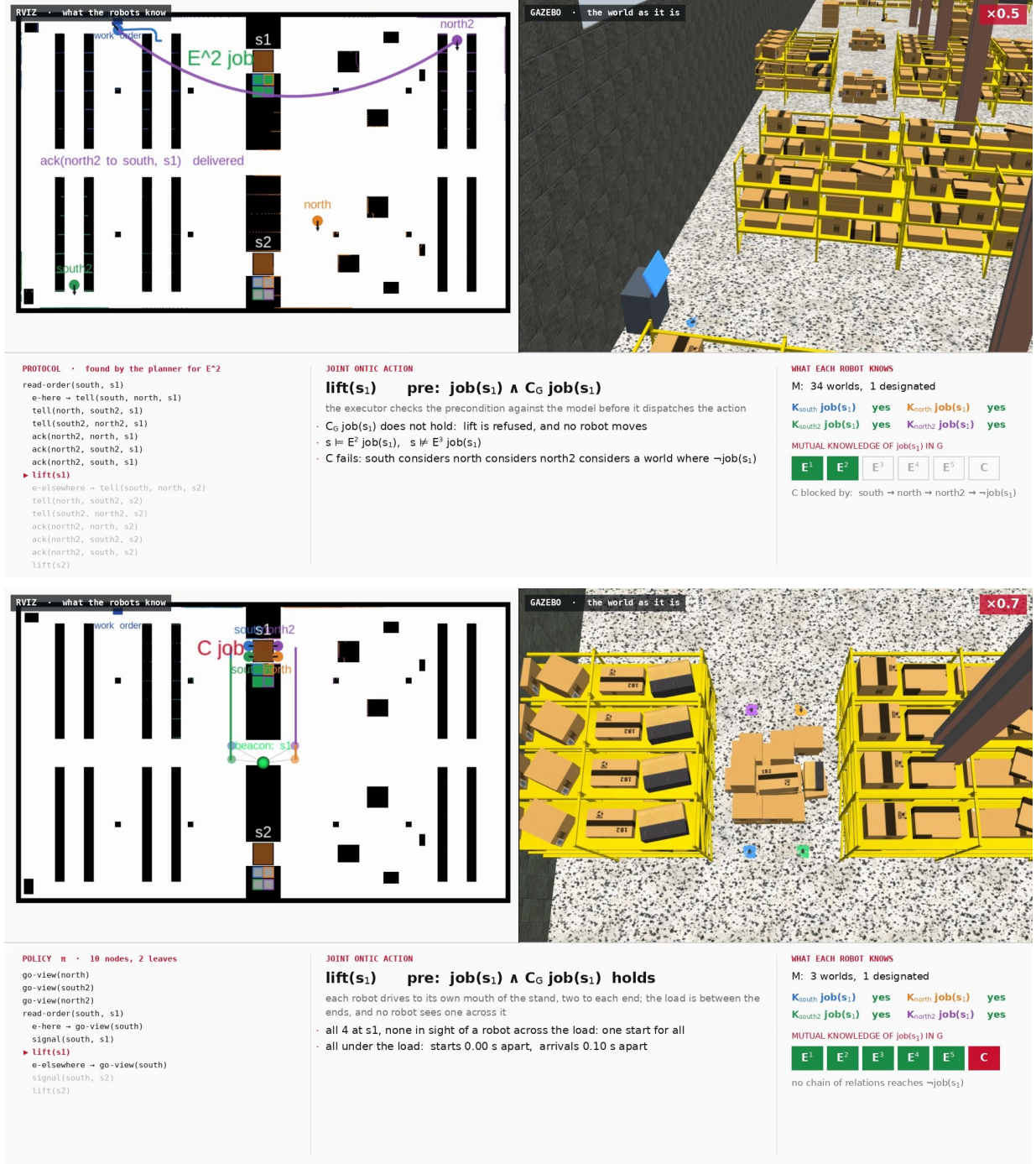


Figure 3: Two frames of the film. Above, the radio floor after the sixth message: 34 worlds, E^2 among the four robots, and the lift refused. Below, the beacon floor during the lift, seen from above the bay: C, and the four robots under the load.

10 Defects

A base that drifts after turning. The first three four-robot runs of the beacon floor, two with the order naming s_1 and one with s_2 , lifted with two robots short of the load and arrivals 11.4 to 11.6 s apart. A simulated Waffle that has turned on the spot and is then driven straight, with an angular command of exactly zero, curves back toward the heading it arrived on, by about 47 degrees over the metre and a half into the bay. A probe in an isolated simulator reproduced it, and showed no drift without the prior turn. The outer two robots ran into the side of the bay. The inner two drifted about 0.9 m as well, and were counted as under the load only because the

lift node measured depth into the bay alone. The lift now creeps along the line from each robot's mouth and steers back onto it, and the four arrive together. The companion report's runs used the open-loop creep along the axis of the bay, where a drift of that size still ends inside it; whether they drifted was not measured, and neither was whether the other demonstrations that use the open-loop creep are affected.

A client that stopped waiting. The first run of the four-robot radio floor reported that the planner returned nothing after 15.0s. The planner was still searching; the mission's planner client had waited its default of 15s. On the companion report's floors the planner answers in under a second, while on the four-robot radio floor it searches its whole budget, and the client is now given that budget and fifteen seconds more.

A count of levels that was not the model. The first version of the ladder tool counted only the highest level each robot had heard from each other, and searched over that count. Random sequences replayed through the product update refuted it: after $tell(south, north)$, $tell(north, south)$ and $ack(south, north)$ the depth is E^3 , where the count gave E^2 . The count missed what a message says beyond its content, and the ladder is now searched by the planner over the domain itself.

The simulator at start. gzserver crashed at startup in one run and in one probe, while the second robot's plugins loaded and before the third was spawned. The run was repeated unchanged.

11 What the runs establish

That the companion report's account of the coordinated attack does not depend on there being two robots. With a type for the robots that neither send nor receive a message, the bound of one level per message holds for any group, and the radio floor has no policy for any number of robots or messages; one signal from viewpoints that are common knowledge yields common knowledge over all of them.

That the account holds on the floor with four robots: six delivered messages among them gave E^2 , computed from the live model after each message as the replay computed it offline, and the executor refused the lift; one signal gave common knowledge over all four, and the four lifted together.

And that what scale changes is the cost. Replanning finds the beacon policy for ten robots in under a second, while the proofs on the radio floor run out of budget from three robots and two levels; and among N robots each level of mutual knowledge cost $N - 1$ messages in every cell the planner settled, because a message carries what it implies about who could have sent it.

12 Directions for the next iteration

The cost of a level. Among N robots, E^k cost $k(N - 1)$ messages in every cell the planner settled, and in the cells it did not settle it showed that fewer do not suffice. For two robots the count is Proposition 2. Whether it holds for every N is open, and needs an argument, since no finite set of cells settles it.

Positions with more robots. The sighting and the unconfirmed announcement of the companion report are written for a pair, and a version for a group of robots was not attempted.

Larger floors. The floor has places for four robots. More would need viewpoints of the beacon and corners under the loads for each, and the planner reaches the beacon policy by replanning for ten.

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Reproduction

```
ros2 launch coordinated_attack_demo coordinated_attack_launch.py robots:=4 messages:=2
ros2 launch coordinated_attack_demo coordinated_attack_launch.py robots:=4 messages:=2 order:=s2
ros2 launch coordinated_attack_demo coordinated_attack_launch.py robots:=4 messages:=2 floor:=radio
bash tools/validate.sh # thirteen checks, no simulator
python3 tools/scaling.py --out /tmp/ca_scaling # the two planning tables
python3 tools/ladder.py --table 2,3,4,5 --depths 1,2,3,4,5 --out /tmp/ca_ladder
python3 tools/ladder.py --replay protocols/radio-n4-m2.json --out /tmp/ca_replay
ROBOTS=4 MESSAGES=2 bash tools/record_demo.sh beacon s1 /tmp/raw_n4.mkv /tmp/run_n4.log
ROBOTS=4 MESSAGES=2 bash tools/record_demo.sh radio s1 /tmp/raw_n4r.mkv /tmp/run_n4r.log
```